

MATH 3670 Homework 3

Saahas Swaroop

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Ch.5: 1,2,3,5,6,7,10(a),11,18,19(a,b),20(a,b)

1

$$P(X \geq 2) = \sum_{k=2}^n \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=2}^4 \binom{4}{k} .6^k .4^{4-k} \\ \binom{4}{2} .6^2 .4^2 + \binom{4}{3} .6^3 .4^1 + \binom{4}{4} .6^4 .4^0 = 0.82$$

2

$$P(X \geq 3) = \sum_{k=3}^5 \binom{5}{k} .2^k (.8)^{5-k} \\ P(X \geq 3) = \binom{5}{3} .2^3 .8^2 + \binom{5}{4} .2^4 .8^1 + \binom{5}{5} .2^5 .8^0 = 0.058$$

3

$$P(X = 7) = \binom{10}{7} .7^7 .3^3 = 0.534$$

5

$$P(X \geq 1) = \sum_{k=1}^2 \binom{2}{k} p^k (1-p)^{2-k} \\ P(X \geq 1) = \binom{2}{1} p(1-p) + \binom{2}{2} p^2 = 2p - p^2 \\ P(X \geq 2) = \sum_{k=2}^4 \binom{4}{k} p^k (1-p)^{4-k} \\ P(X \geq 2) = \binom{4}{2} p^2 (1-p)^2 + \binom{4}{3} p^3 (1-p) + \binom{4}{4} p^4 = 6p^2(1-p)^2 + 4p^3(1-p) + p^4 = 3p^4 + 2p^3 + 6p^2 \\ 2p - p^2 \leq 3p^4 + 2p^3 + 6p^2$$

$$p \geq 0.27$$

$$P(X \geq 1) = \binom{2}{1}p(1-p) + \binom{2}{2}p^2 = 2p - p^2$$

6

$$E(X) = np = 7$$

$$\text{Var}(X) = np(1-p) = 2.1$$

$$1-p = 0.3, p = 0.7, n = 10$$

a

$$P(X = 4) = \binom{10}{4}.7^4.3^6 = 0.0368$$

b

$$P(X > 12) = 0$$

7

a

$$P(X \leq i) = \sum_{k=0}^i \binom{n}{k} p^k (1-p)^{n-k}$$

$$P(X \geq n-i) = \sum_{k=n-i}^n \binom{n}{k} (1-p)^{n-k} p^k$$

Both probabilities will be equal due to the nature of their summation. The former will begin with $\binom{n}{0}$ and sum up to $\binom{n}{i}$. The latter will begin with $\binom{n}{n-i}$ and sum up to nn . However, we will note that this is simply the former summation but in reverse, as $\binom{n}{0} = \binom{n}{n}$ and $\binom{n}{i} = \binom{n}{n-i}$.

b

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} \cdot p^k \cdot (1-p)^{n-k}$$

$$P(Y = n - k) = \binom{n}{n-k} (1-p)^{n-k} p^{n-(n-k)} = \frac{n!}{(n-k)!(k)!} \cdot p^k \cdot (1-p)^{n-k}$$

1 10

a

$$P(X = 2) = \binom{10}{2} \cdot .1^2 \cdot .9^8 = 0.194$$

$$P(X = 2) = \frac{e^{-1} 1^2}{2!} = 0.184$$

11

a

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{e^{-.5} .5^0}{0!} = 0.393$$

b

$$P(X = 0) = e^{-0.5} \cdot \frac{0.5^1}{1!} = 0.303$$

c

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-0.5} \cdot \frac{0.5^0}{0!} - e^{-0.5} \cdot \frac{0.5^1}{1!} = 0.09$$

18

$$P(\text{kept}) = P(0) + P(1) = \frac{\binom{80}{10} + 20\binom{80}{9}}{\binom{100}{10}} = 0.0296$$